

# Column Description

BAP Service Time Prediction — Puerto de San Antonio, Chile

Dataset: training\_dataset.csv | 5,597 rows | 44 columns

## 1. Target Variables / Variables Objetivo

Two service time metrics computed from timestamps. Not derived from Excel formulas.

| Column                 | Spanish          | Description                                                     | Calculation                                                   |
|------------------------|------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| estadia_sitio_hours    | Estadía sitio    | Berth occupation time (hours). First mooring to last unmooring. | $\text{Código. espía desatraque} - 1\text{era espía atraque}$ |
| tiempo_en_puerto_hours | Tiempo en puerto | Time inside port (hours). Pilot boarding to departure.          | $\text{Zarpe} - \text{Fecha práctico atraque}$                |

**Note:** F. arribo records arrival at the anchorage (outside the port), not port entry. Pilot boarding marks actual port entry. Dry bulk vessels wait a median of 207h (~8.6 days) at anchor.

## 2. Direct Features / Características Directas

Pre-berthing information only. No departure-time data or berth assignment (BAP decision variable).

| Column              | Spanish Original        | Description                                    | Calculation                            |
|---------------------|-------------------------|------------------------------------------------|----------------------------------------|
| vessel_code         | Cód. nave               | Unique vessel identifier (IMO)                 | Direct                                 |
| vessel_name         | Nave                    | Vessel name / Nombre de la nave                | Direct                                 |
| vessel_type         | T. nave                 | Original vessel type (16 categories)           | Direct                                 |
| vessel_type_group   | T. nave (agrupado)      | Aggregated vessel type (7 groups)              | Mapped from vessel_type                |
| trg                 | TRG                     | Gross registered tonnage / Tonelaje bruto      | Direct                                 |
| terminal            | Terminal                | Port terminal (DP World, STI, PANUL, QC, EPSA) | Direct                                 |
| agency              | Agencia                 | Maritime agency name / Agencia marítima        | Split "RUT - NAME" on " - ", take name |
| shipping_line       | L. naviera              | Shipping line. "UNKNOWN" if missing (30%)      | Direct, NaN → UNKNOWN                  |
| service_route       | Servicio                | Service route. "UNKNOWN" if missing (30%)      | Direct, NaN → UNKNOWN                  |
| origin_port         | Puerto origen           | Port of origin / Puerto de origen              | Direct                                 |
| dest_port           | Puerto destino          | Port of destination / Puerto de destino        | Direct                                 |
| origin_region       | Puerto origen (región)  | Geographic region of origin (14 regions)       | Mapped via port_regions.py             |
| dest_region         | Puerto destino (región) | Geographic region of destination               | Mapped via port_regions.py             |
| draft_arrival_bow   | C. arribo proa          | Bow draft on arrival (meters)                  | Direct                                 |
| draft_arrival_stern | C. arribo popa          | Stern draft on arrival (meters)                | Direct                                 |
| draft_arrival_mean  | Calado medio arribo     | Mean arrival draft (meters)                    | $(\text{proa} + \text{popa}) / 2$      |
| max_arrival_draft   | Calado máximo arribo    | Maximum arrival draft (meters)                 | $\max(\text{proa}, \text{popa})$       |

| Column             | Spanish Original  | Description                              | Calculation |
|--------------------|-------------------|------------------------------------------|-------------|
| draft_trim_arrival | Trimado al arribo | Arrival trim (m). Positive = stern-heavy | popa - proa |

### 3. Temporal Features / Características Temporales

Derived from F. arribo (arrival at anchorage datetime).

| Column              | Spanish          | Description                | Calculation           |
|---------------------|------------------|----------------------------|-----------------------|
| arrival_month       | Mes de arribo    | Month of arrival (1-12)    | F. arribo.month       |
| arrival_day_of_week | Día de la semana | Day of week (0=Mon, 6=Sun) | F. arribo.weekday()   |
| arrival_hour        | Hora de arribo   | Hour of arrival (0-23)     | F. arribo.hour        |
| arrival_year        | Año de arribo    | Year of arrival            | F. arribo.year        |
| quarter             | Trimestre        | Quarter of the year (1-4)  | (month - 1) // 3 + 1  |
| is_weekend_arrival  | Fin de semana    | Weekend arrival flag (0/1) | 1 if day_of_week >= 5 |

### 4. Historical Vessel Features / Características Históricas

Expanding-window statistics over prior visits of the same vessel, sorted chronologically by first mooring time. Shifted by 1 to exclude the current visit (no data leakage). NaN for first visits (1,550 vessels).

| Column                            | Spanish                   | Calculation                                                                 |
|-----------------------------------|---------------------------|-----------------------------------------------------------------------------|
| vessel_avg_berth_stay             | Promedio estadía nave     | groupby(vessel_code)[estadia_sitio].expanding().mean().shift(1)             |
| vessel_median_berth_stay          | Mediana estadía nave      | groupby(vessel_code)[estadia_sitio].expanding().median().shift(1)           |
| vessel_std_berth_stay             | Desv. est. estadía nave   | groupby(vessel_code)[estadia_sitio].expanding().std().shift(1)              |
| vessel_visit_count                | Visitas previas nave      | groupby(vessel_code).cumcount() (0 = first visit)                           |
| vessel_last_berth_stay            | Última estadía nave       | groupby(vessel_code)[estadia_sitio].shift(1)                                |
| vessel_avg_berth_stay_at_terminal | Promedio nave en terminal | groupby([vessel_code, terminal])[estadia_sitio].expanding().mean().shift(1) |
| vessel_visit_count_at_terminal    | Visitas nave en terminal  | groupby([vessel_code, terminal]).cumcount()                                 |

### 5. Group-Level Features / Características de Grupo

Expanding-window averages at higher grouping levels, shifted by 1. Useful as fallback predictions when vessel-specific history is unavailable.

| Column                 | Spanish                | Calculation                                                                       |
|------------------------|------------------------|-----------------------------------------------------------------------------------|
| type_terminal_avg_stay | Promedio tipo+terminal | groupby([vessel_type_group, terminal])[estadia_sitio].expanding().mean().shift(1) |
| type_avg_stay          | Promedio tipo          | groupby(vessel_type_group)[estadia_sitio].expanding().mean().shift(1)             |
| terminal_avg_stay      | Promedio terminal      | groupby(terminal)[estadia_sitio].expanding().mean().shift(1)                      |

### 6. Reference Columns / Columnas de Referencia

Not intended as model features. Included for traceability and analysis.

| Column                  | Spanish Original        | Description                            | Calculation                        |
|-------------------------|-------------------------|----------------------------------------|------------------------------------|
| arrival_datetime        | F. arribo               | Arrival at anchorage                   | Direct                             |
| pilot_boarding_datetime | Fecha práctico atraque  | Pilot boarding (port entry)            | Direct                             |
| first_mooring_datetime  | 1era espía atraque      | First mooring line secured             | Direct                             |
| last_unmooring_datetime | Última espía desatraque | Last unmooring line cast off           | Direct                             |
| departure_datetime      | Zarpe                   | Vessel departure                       | Direct                             |
| espera_preatraque_hours | Espera pre-atraque      | Anchorage wait time (hours)            | (Zarpe - F.arribo) - estadia_sitio |
| quality_flag            | Bandera calidad         | 1 if dispatch < reception (12 records) | Timestamp comparison               |
| split                   | Partición               | train / validation / test              | Based on arrival date thresholds   |

## 7. Vessel Type Aggregation / Agrupación de Tipos

| Group           | Original Types / Tipos Originales                            | Count | Avg Berth (h) |
|-----------------|--------------------------------------------------------------|-------|---------------|
| Container       | Contenedor                                                   | 3,066 | 36            |
| Dry Bulk        | Carga Seca Granel, Mineral/Granel/Petrolero                  | 951   | 74            |
| Vehicle Carrier | Autero, Autotrasbordo                                        | 749   | 36            |
| Liquid Bulk     | Transp. Quimico, Transp. Liquido, Transp. Asfalto, Petrolero | 587   | 22            |
| General Cargo   | Tradicional, Carga de Proyecto, Chipero, Refrigerado, Otros  | 171   | 36            |
| Passenger       | Pasajeros                                                    | 80    | 19            |
| Other           | Nave Armada                                                  | 1     | 65            |

## 8. Port Regions / Regiones Portuarias

| Región / Region  | Examples / Ejemplos                                     |
|------------------|---------------------------------------------------------|
| Chilean_North    | Arica, Iquique, Antofagasta, Mejillones, Coquimbo       |
| Chilean_Central  | Valparaiso, San Antonio, Coronel, Lirquen, San Vicente  |
| Chilean_South    | Puerto Montt, Punta Arenas, Chacabuco                   |
| Peru             | Callao, Ilo, Matarani, Paita                            |
| Brazil           | Santos, Rio Grande, Paranagua, Rio de Janeiro           |
| Argentina        | Buenos Aires, Rosario, Zarate, Bahia Blanca, Montevideo |
| Ecuador_Colombia | Guayaquil, Buenaventura, Manta                          |
| Central_America  | Balboa, Cristobal, Manzanillo, Lazaro Cardenas          |
| North_America    | Houston, Long Beach, Los Angeles, Vancouver             |
| Asia             | Shanghai, Busan, Singapore, Hong Kong, Yokohama         |
| Europe           | Rotterdam, Zeebrugge, London, Livorno                   |
| Africa           | Durban, Capetown, Port Elizabeth                        |
| Oceania          | Melbourne, Fremantle, Tauranga                          |
| Other            | Unmapped ports / Puertos no mapeados                    |